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AI powered Policy-Grounded Agentic AI for Claim Readiness and Appeal Support

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Introduction

- Complex insurance policies make it difficult for policyholders to understand coverage, claim requirements, and claim decisions.
- Incomplete or inconsistent documentation can result in claim queries, processing delays, or avoidable rejections.
- After a claim denial, policyholders may struggle to understand the reason, determine whether an appeal may be appropriate, and identify the supporting evidence required.
- Existing support systems and generic AI tools provide limited policy-specific guidance and lack comprehensive assistance across the claim and appeal process.
- A policy-grounded Multi-Agent Agentic AI approach can address this gap by supporting claim readiness assessment, denial analysis, evidence identification, and appeal preparation.

Motivation

- Empower policyholders to understand their insurance policies, claim requirements, and claim decisions.
- Reduce avoidable claim rejections by identifying missing or inconsistent documentation before claim submission.
- Support informed appeals by analyzing whether a denial appears consistent with the policy and identifying relevant supporting evidence.
- Provide accessible, policy-specific assistance beyond traditional agents, customer support, and generic AI tools.
- Ensure human oversight in high-stakes decisions, keeping the policyholder involved in reviewing and making final decisions.

Literature Survey

Author, Publisher, Year	Title	Methodology	Advantages	Limitations
S. Hanmante et al.; Elsevier, *Software Impacts*; 2026	InsuranceRAG: A Multi-Module Retrieval-Augmented AI System for Intelligent Insurance Assistance	Uses RAG, embeddings, FAISS, and LLMs for insurance policy retrieval, policy recommendation, and conversational assistance. Relevant policy clauses are retrieved to provide policy-grounded responses.	Improves policy-specific responses through retrieval; supports clause-level information retrieval; provides a modular insurance AI architecture	Primarily focuses on policy understanding and recommendation; does not address claim readiness, denial analysis, evidence validation, or appeal generation.
Jin Liu et al.; Association for Computational Linguistics (ACL); 2026	InsLogicBench: An Argumentation Logic Grounded Benchmark for Complex Insurance Claims Adjudication	Introduces a benchmark for evaluating LLMs on complex insurance claim adjudication using policy clauses, claim facts, and logical reasoning to assess whether models can perform policy-based reasoning.	Enables systematic evaluation of LLM reasoning over insurance policies. Supports explainable and logic-based claim analysis beyond simple information retrieval.	Primarily provides a benchmark for evaluation rather than a complete insurance assistance system. It does not implement an end-to-end claim or appeal workflow.

Author, Publisher, Year	Title	Methodology	Advantages	Limitations
M. Fink et al.; Elsevier, *Academic Radiology*; 2026	Decreasing Administrative Effort Related to Non-Approval of Image-guided Procedures Using Large Language Models – The DENIED-AI Pilot Study	Evaluates multiple LLMs using zero-shot, few-shot, and RAG approaches for generating insurance denial appeal letters. Generated appeals are evaluated by medical experts for accuracy, personalization, readability, and usability.	Demonstrates that LLMs can generate useful appeal drafts and reduce administrative effort. It also compares multiple LLM approaches with expert human evaluation.	Focuses on image-guided radiology procedures and simulated cases. Hallucinations and fabricated references were observed, requiring human review.
Timothy Owolabi; Oxford University Press, *JAMIA Open*; 2025	Transforming Appeal Decisions: Machine Learning Triage for Hospital Admission Denials	Uses machine learning classifiers trained on historical hospital admission denial cases to identify denials that are more likely to be overturned through appeal.	Uses real-world denial data and demonstrates the potential of ML for prioritizing cases for appeal. Provides quantitative evaluation of appeal decision support.	Limited to hospital admission denials and focuses on predicting appeal outcomes. It does not provide policy-grounded explanations, document verification, or appeal generation.

Research Gap

- **Limited Claim Readiness Assessment:** Existing systems provide limited automated validation of claims before submission.
- **Limited Denial and Evidence Analysis:** Existing approaches do not jointly analyze denial reasons, policy clauses, and supporting medical evidence.
- **Limited End-to-End Insurance Support:** Current solutions address individual tasks rather than integrating claim preparation, denial analysis, and appeal support.
- **Limited Agentic Appeal Assistance:** Existing systems provide limited stateful, human-in-the-loop workflows for personalized and policy-grounded appeal generation.

Problem Statement

Insurance claim processing can be difficult due to complex policies, incomplete documentation, and technical claim decisions. Existing solutions often address individual tasks such as policy assistance, claim verification, or appeal generation. There is a need for a policy-grounded, multi-agent approach that supports the claim lifecycle from claim readiness and denial analysis to evidence identification and appeal support, while ensuring reliable and human-supervised decisions.

Key Challenges:

- Understanding complex insurance policies, coverage conditions, exclusions, and claim requirements.
- Extracting and validating relevant information from policies, denial notices, and supporting documents.
- Determining claim readiness, identifying missing evidence, and assessing whether a denial is consistent with the policy.
- Coordinating multiple AI agents while ensuring policy-grounded reasoning, reliable outputs, and Human-in-the-Loop (HITL) oversight.

Objectives

- Develop a Policy-Grounded AI System: Build a Multi-Agent AI system using Python, LangGraph, LangChain, MCP, and RAG to analyze insurance policies and support claim and appeal processes.

- Automate Document & Claim Validation: Use OCR, document classification, duplicate detection, and rule-based validation to identify missing or inconsistent claim documents.

- Assess Claim Readiness: Apply LLM-based medical and policy analysis to detect exclusions, waiting periods, financial limits, and medical inconsistencies before claim submission.

- Analyze Claim Denials: Use RAG-based policy and evidence analysis to verify denial reasons and clauses against the actual policy and identify missing evidence or insurer errors.

- Support Appeal Preparation: Generate evidence-backed appeal drafts using relevant policy clauses and medical evidence, with human approval, feedback, and escalation workflows through LangGraph.

References

[1] S. Hanmante, S. Patil, and A. K. Shahade, “A Multi Module A.I. System for Intelligent Health Insurance Support Using Retrieval Augmented Generation,” *Scientific Reports*, vol. 16, 2026.

[2] M. Fink, C. J. McCarthy, et al., “Decreasing Administrative Effort Related to Non-Approval of Image-guided Procedures Using Large Language Models – The DENIED-AI Pilot Study,” *Academic Radiology*, vol. 33, no. 8, pp. 3221–3231, 2026.

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[4] T. Owolabi, “Transforming Appeal Decisions: Machine Learning Triage for Hospital Admission Denials,” *JAMIA Open*, vol. 8, 2025.